

Measured Usage Guide

Scientific Data Analysis v2.6

Reproducible observation and evidence-based optimization

Operational health: benchmark matrix, local harness, mother/child comparison, and
app-readiness gates

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Translation of the preserved private v2.6 guide at commit `d6583f1`. Original TeX SHA-256:
`86bf78142300618242b3036d4537e88009ac353d30b96a798d4c2d66a562c131`. Private source:
`skills/scientific-data-analysis/.cache/active_references/guides/guia_scientific_`
`data_analysis_v2_6.tex`.

Recorded results and closure steps are historical, not current verification. This release guide does not
synchronize or modify Scientific Workbench; the app remains outside its scope. Historical commands
are retained for traceability: use fresh output directories when reproducing them, without
overwriting earlier evidence. The original remains unchanged privately.

Abstract

`scientific-data-analysis` v2.6 continues the modular architecture stabilized in v2.5, with a different question. v2.5 reduced static loading and closed required plugin-eval fixes. v2.6 asks where representative usage actually incurs cost. It introduces a benchmark matrix, an observed-usage harness, reproducible `observed_usage.jsonl` results, a mother/child comparison, and a conservative policy: optimize only when measurements justify the change.

The measurement is a local proxy, not a reading of actual weekly Codex tokens. v2.6 therefore promises no reduction in account usage limits. It provides verifiable infrastructure for measuring and comparing future decisions while preserving v2.5 compatibility.

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1. Executive overview

v2.6 focuses on measurement and restraint. It adds no capabilities, changes no scientific contracts, and leaves Scientific Workbench untouched. It makes efficiency discussions evidence-based through reproducible scenarios, parseable outputs, and an initial ranking of local cost.

The product decisions are:

- `scientific-data-analysis` remains the normal entrypoint for ambiguous or mixed tasks.
- Use `scientific-data-astro`, `scientific-data-documents`, `scientific-data-notebooks`, or `scientific-data-maintainer` when the intent is already specific.
- Results use `token_measurement_kind=local_proxy`; they are not actual account token counts.
- Registries, artifact types, run bundles, app hints, and app-readiness contracts remain unchanged.
- Optimization is limited to stabilizing the harness and documenting measurable decisions.

2. From v2.5 to v2.6

Area	v2.6 change
Charter	<code>v2-6-measured-usage-charter.md</code> defines objectives, non-goals, and closure criteria.
Observed contract	<code>v2-6-observed-usage-contract.md</code> separates local proxy measurement from actual tokens.
Benchmark matrix	<code>v2-6-benchmark-matrix.json</code> contains 8 synthetic or anonymized scenarios.
Harness	<code>audit_v2_6_observed_usage_harness.py</code> runs scenarios and saves <code>observed_usage.jsonl</code> .
Regressions	Adds v2.6 checks for baseline, matrix, observed usage, and app-readiness sanity.
Router comparison	<code>v2-6-routing-cost-findings.md</code> documents mother versus child routes.
Optimization	<code>v2-6-cost-optimization-log.md</code> records decisions; no large refactor or public trigger change.
Maintainer scenario	Uses a maintainer-specific regression and captures JSON stdout as a parseable summary.
Guide	Replaces v2.5 as the active release document.

3. What remains unchanged

- No new capability, public path deletion, or public path rename.
- No edits to Scientific Workbench.
- No synchronization of the installed family until the final phase.
- No forced restructuring of mother and children.
- No claim of actual token savings without comparable actual measurements.
- No changes to the scientific contracts of astro, documents, notebooks, or maintainer routes.

4. Measured scenarios

The v2.6 matrix covers eight scenarios. All inputs are synthetic or anonymized; originals are not modified.

Scenario	Owner, route, and input
table_professional_csv	notebooks; direct child; synthetic professional CSV
mixed_folder_router	mother; router; synthetic mixed folder
legacy_notebook_inspect	notebooks; direct child; synthetic legacy notebook
document_admin_intake	documents; direct child; administrative document package
fits_simple_inspect	astro; direct child; minimal synthetic FITS
optional_stilts_block	astro; optional; simulated absent STILTS backend
maintainer_surface_check	maintainer; maintainer route; v2.5 maintainer regression
router_vs_direct_table	mother; router; the same CSV used for direct profiling

5. Observed result

Reference execution:

```
python scripts/audit_v2_6_observed_usage_harness.py \
  --output-dir tmp/v2_6_phase2_observed_usage_harness
```

Scenario	Status	Proxy	ms
table_professional_csv	PASS	1975	459
mixed_folder_router	PASS	20484	94
legacy_notebook_inspect	Controlled block	811	67
document_admin_intake	PASS	3605	602
fits_simple_inspect	PASS	1313	110
optional_stilts_block	Controlled block	1457	73
maintainer_surface_check	PASS	162	44
router_vs_direct_table	PASS	20565	90

“Proxy” is the local token proxy; “ms” is duration in milliseconds. “Controlled block” preserves the original machine status `BLOCKED_CONTROLADO`.

Interpretation:

- There are 6 PASS results and 2 controlled blocks.
- The notebook stops cleanly under system Python because `nbformat` is missing.
- The absent STILTS backend blocks cleanly; this is not treated as a core error.
- The CSV comparison shows a heavier mother route than the direct child when the goal is already specific.
- `local_proxy` supports internal comparisons, not promises of actual weekly savings.

6. Mother versus child routing

The mother remains the appropriate entrypoint for broad requests such as “inspect this folder” or “help me with this project.” Direct children provide a shorter route when the intent is clear:

- FITS, WCS, photometry, spectra, or astronomy: `scientific-data-astro`.
- PDF, DOCX, LaTeX, presentations, or reporting: `scientific-data-documents`.
- CSV, notebooks, containers, time series, or tables: `scientific-data-notebooks`.
- Release gates, plugin-eval, or maintenance: `scientific-data-maintainer`.

v2.6 does not change trigger text or public contracts on the basis of one measurement. It documents evidence and provides repeatable regression checks.

7. v2.6 regressions

Main regressions:

```
python scripts/audit_v2_6_phase0_baseline_regression.py
python scripts/audit_v2_6_benchmark_matrix_regression.py
python scripts/audit_v2_6_observed_usage_harness.py --output-dir tmp/
    v2_6_phase2_observed_usage_harness
python scripts/audit_v2_6_observed_usage_regression.py --output-dir tmp/
    v2_6_phase2_observed_usage_harness
python scripts/audit_v2_6_app_ready_cost_sanity_regression.py --output-dir tmp/
    v2_6_phase5_app_ready_sanity
```

Inherited gates that must continue to pass:

```
python scripts/audit_v2_5_budget_deferral_regression.py
python scripts/audit_v2_4_budget_architecture_regression.py
python scripts/sync_public_surface_docs.py --check
python scripts/portable_smoke_test.py --profile core
```

8. Historical closure criteria

The original v2.6 closure required all ten conditions:

1. The benchmark matrix covers all 5 skills.
2. The harness produces `observed_usage.jsonl`, `result.json`, and `summary.md`.
3. All 8 scenarios are PASS, WARNING, or justified BLOCKED_CONTROLADO, with no FAIL/ROTO.
4. App-readiness sanity validates summaries, artifacts, statuses, and `original_modified=false`.
5. `guia_scientific_data_analysis_v2_6.tex` compiles to PDF.
6. The PDF and TeX are copied to the Desktop, as required by that historical local handoff.
7. v2.6 and inherited gates pass in the editable family.
8. Installed synchronization occurs only after green gates.
9. Post-synchronization gates pass.
10. `evaluate-skill` runs against the installed mother and children.

9. Recommended usage after v2.6

For broad tasks:

```
python scripts/scientific_workflow_router.py plan input --task "..."
```

For specific tasks:

```
python scripts/profile_table.py data.csv --summary-json summary.json
python scripts/inspect_fits.py image.fits --summary-json summary.json
python scripts/document_intake_workbench.py docs/ --summary-json summary.json
```

To repeat the v2.6 measurement:

```
python scripts/audit_v2_6_observed_usage_harness.py --output-dir tmp/v2_6_observed_usage
python scripts/audit_v2_6_observed_usage_regression.py --output-dir tmp/
    v2_6_observed_usage
```

10. Plugin-eval evaluation update

After conservative hardening following v2.6, the mother and four children were optimized for `plugin-eval:evaluate-skill` without removing capabilities or breaking public paths. The full content was deferred: guides, long references, historical duplicates, examples, and heavy helpers remained available outside the initial content Codex reads when invoking a skill.

The remaining informational item was `coverage-artifacts-unavailable`. To address it, `audit_v2_7_plugin_eval_coverage_artifacts.py` generates `coverage-summary.json` for each skill. It reports smoke/regression coverage observed with `trace`, not exhaustive project coverage. It gives `plugin-eval` a real, traceable coverage signal instead of an empty field.

Recorded editable-family state before synchronization:

Skill	Rating	Deductions
scientific-data-analysis	100/A	none
scientific-data-astro	100/A	none
scientific-data-documents	100/A	none
scientific-data-notebooks	100/A	none
scientific-data-maintainer	100/A	none

Three invariants must remain:

- Public paths continue to exist, even when they point to deferred content.
- Coverage artifacts are small, parseable, and regenerable.
- `evaluate-skill` runs on mother and children after every synchronization.

11. Conclusion

v2.6 provides reproducible evidence for reasoning about cost, preserves the v2.5 architecture, and avoids unnecessary contract changes. Use the mother for ambiguity, children for defined tasks, and require benchmark evidence before future optimizations.